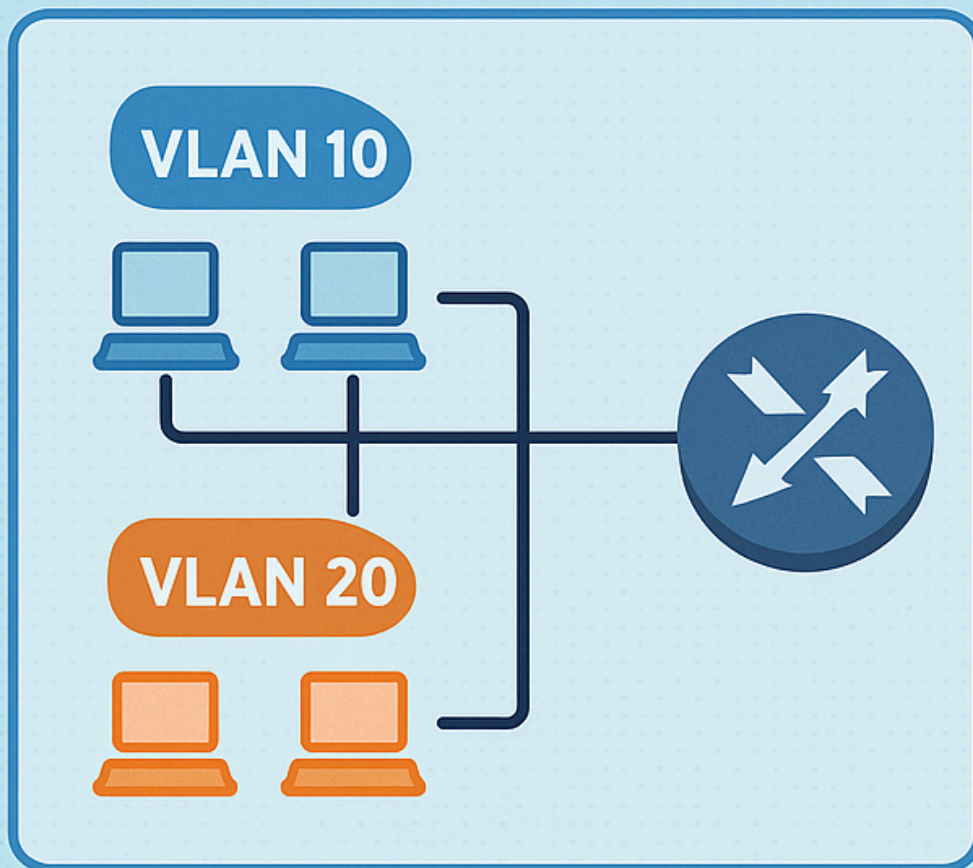


ROUTER-ON-A-STICK

INTER-VLAN ROUTING



Prepared By: Hosni Zaaraoui

What is it?

Router-on-a-Stick is a network configuration used to enable communication between VLANs using a single physical router interface. The router uses subinterfaces (logical divisions of a physical interface) to route traffic between multiple VLANs over a trunk link.



Why use it?

In a network where multiple VLANs are used to segment traffic (e.g., Sales, HR, IT), devices in different VLANs can't communicate by default. Router-on-a-Stick solves this by allowing inter-VLAN routing without needing multiple physical interfaces.

How it works:

- 1. A trunk link connects the switch to the router.**
- 2. The router interface is split into subinterfaces (e.g., Gig0/1.10, Gig0/1.20).**
- 3. Each subinterface is assigned to a VLAN and given an IP address (gateway for that VLAN).**
- 4. The router performs Layer 3 routing between VLANs.**

Example:



```
interface GigabitEthernet0/1.10  
  encapsulation dot1Q 10  
  ip address 192.168.10.1  
  255.255.255.0
```

```
interface GigabitEthernet0/1.20  
  encapsulation dot1Q 20  
  ip address 192.168.20.1  
  255.255.255.0
```